

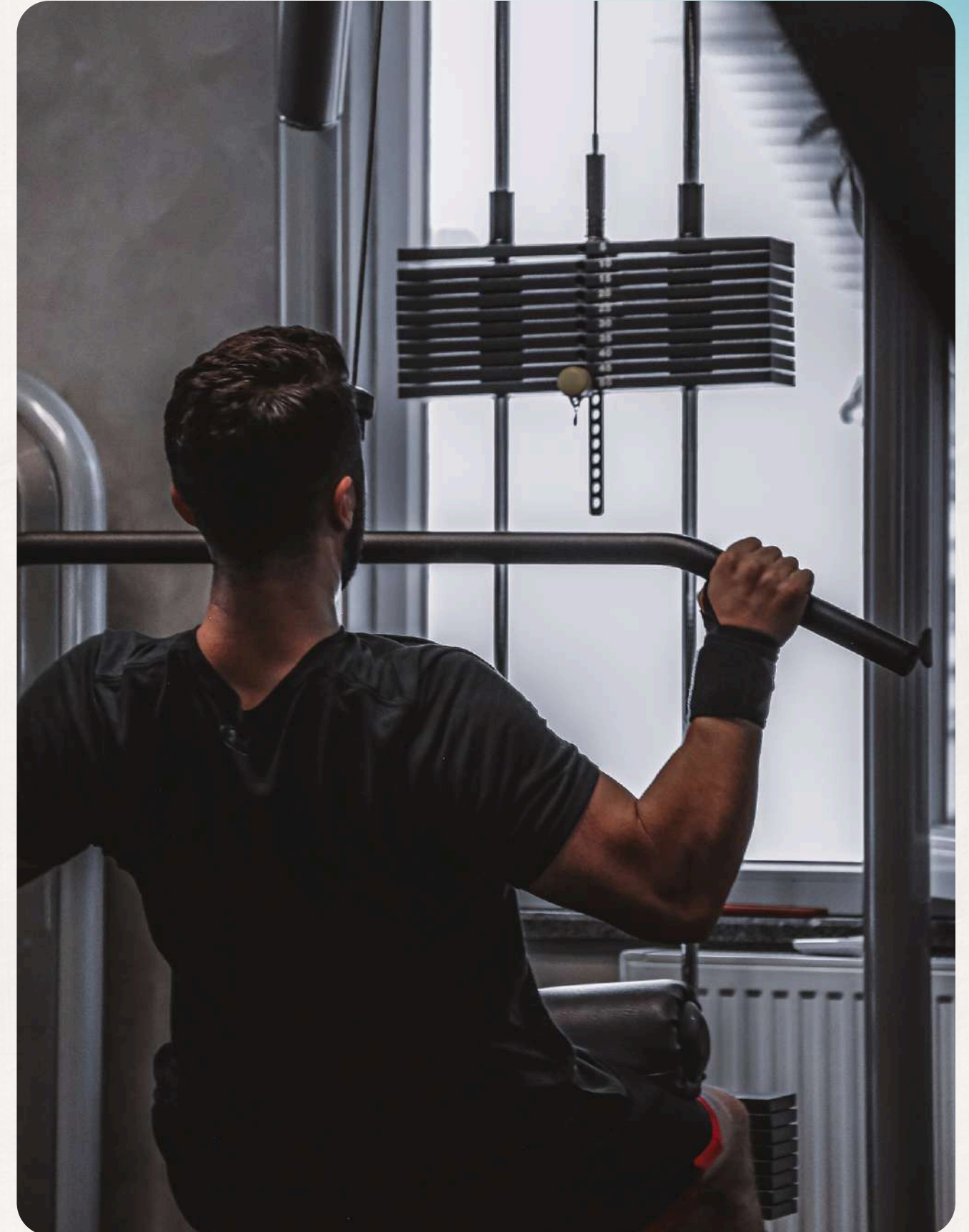
MADE BY~
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GYM & FITNESS MEMBERSHIP MANAGEMENT SYSTEM

A ROLE-BASED PLATFORM THAT DIGITIZES MEMBERSHIPS, BOOKINGS,
ATTENDANCE, COACHING, PAYMENTS, AND GYM OPERATIONS.

1. PROBLEM STATEMENT

- Gyms manage registrations, class schedules, PT bookings, attendance, equipment, and billing across disconnected manual processes.
- Registers and spreadsheets are slow, error-prone, and provide limited real-time visibility to branch administrators.
- Members need a simple way to manage plans, bookings, entry passes, and progress.
- Trainers need a dedicated workspace for clients, routines, PT sessions, earnings, and ratings.
- GymLandT brings these workflows together in one role-based web application.



2. OBJECTIVES

Build a responsive, role-based gym management application using Angular.

Automate attendance and class/PT booking tracking.

Simplify membership and personal-training-pack purchases through simulated checkout.

Give trainers a hub for clients, workout/diet routines, sessions, earnings, and ratings.

Give branch admins financial, CRM, equipment, maintenance, and class-management tools.

Use modular Angular architecture with a RESTful Node.js/Express backend.

3. USERS & CORE WORKFLOW

Three roles share one platform, with access controlled by
JWT + role authorization.

01 MEMBER

Plans • Classes • QR Pass
Progress • Ratings

02 TRAINER

Clients • PT Sessions
Workout/Diet • Earnings

03 BRANCH ADMIN

CRM • Finance • Equipment
Classes • QR Verification

4. SYSTEM ARCHITECTURE



01. Angular Frontend

Components • Services
Router • Reactive Forms

02. Node.js + Express

REST Routes
Controllers + Middleware

03. Mongoose

Schemas • ODM
Data Access

04. MongoDB

Collections
Persistent Data

Security layer

protect middleware verifies JWT tokens, while `requireRole` restricts routes to roles such as `BRANCH ADMIN`.

This keeps authentication and authorization centralized and consistent across protected APIs.

5. TECHNOLOGY STACK

Frontend

Angular 17 • TypeScript •
HTML5/CSS3 • RxJS

Backend

Node.js • Express.js • REST
API

Testing

Postman collection + helper
scripts

Development

Angular CLI • Visual Studio
Code

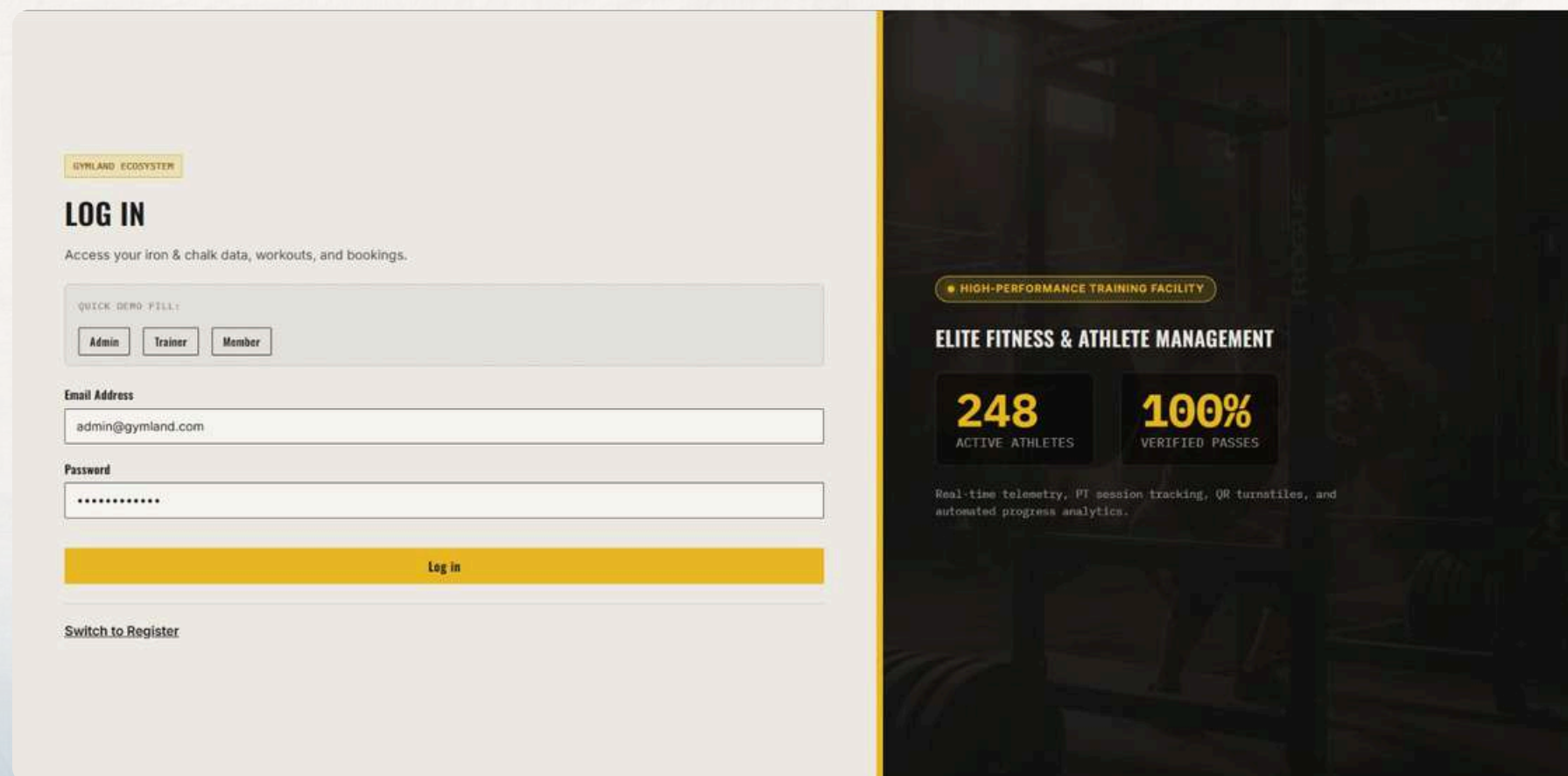
Database

MongoDB • Mongoose

Security

JWT authentication • Role-
based authorization

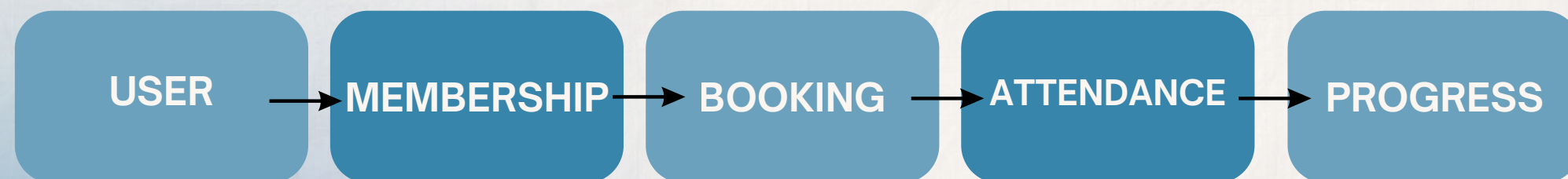
6. FUNCTIONAL MODULES



- Authentication - login/register with role-aware access
- Dashboard - role-aware landing page and navigation hub
- Classes & PT - schedules, bookings, capacity, ratings
- Plans & Shop - memberships and PT packs with dummy checkout
- My QR Pass - scannable member entry pass
- Progress & Routine - body metrics and assigned routines
- Coaching Hub - clients, PT sessions, workout/diet builder, earnings
- Admin Hub - finance, CRM, equipment/maintenance, class creator
- Front Desk QR Scanner - turnstile-style pass verification

7. DATABASE & DATA MODEL

- User • Trainer • Class • Booking • Attendance
- Equipment • Membership • Membership Plan • Workout Plan
- Progress Log • Transaction • Review • Notification • Waitlist



ADMIN HUB

NS & FINANCIAL LEDGER

Inventory, user management, and turnstile verification.

TRAINER CRM

EQUIPMENT & MAINTENANCE

CLASS CREATOR

DIRECTORY

PHONE	ROLE	ASSIGNED TRAINER	PT CREDI
@gym.com	BRANCH ADMIN	-- None --	0
er1@gym.com	TRAINER	-- None --	0
er2@gym.com	TRAINER	-- None --	0
r3@gym.com	MEMBER	Trainer Two	0
r1@gym.com	MEMBER	Trainer One	0
r2@gym.com	MEMBER	Trainer One	0
r4@gym.com	MEMBER	-- None --	0

8. KEY FUNCTIONAL WORKFLOWS

- **Member:** Browse plans → Buy via dummy gateway → Book class → Scan QR pass → Log progress.
- **Trainer:** View assigned clients → Log PT session → Deduct PT credits → Build workout/diet routine → Review earnings.
- **Admin:** Manage CRM → Promote/demote roles → Assign trainer → Track equipment → Create classes → Verify entry.

Strength & Conditioning Specialist // Hypertrophy & Kinetic Rehabilitation

ASSIGNED CLIENTS

WORKOUT & DIET BUILDER

EARNINGS & RATINGS

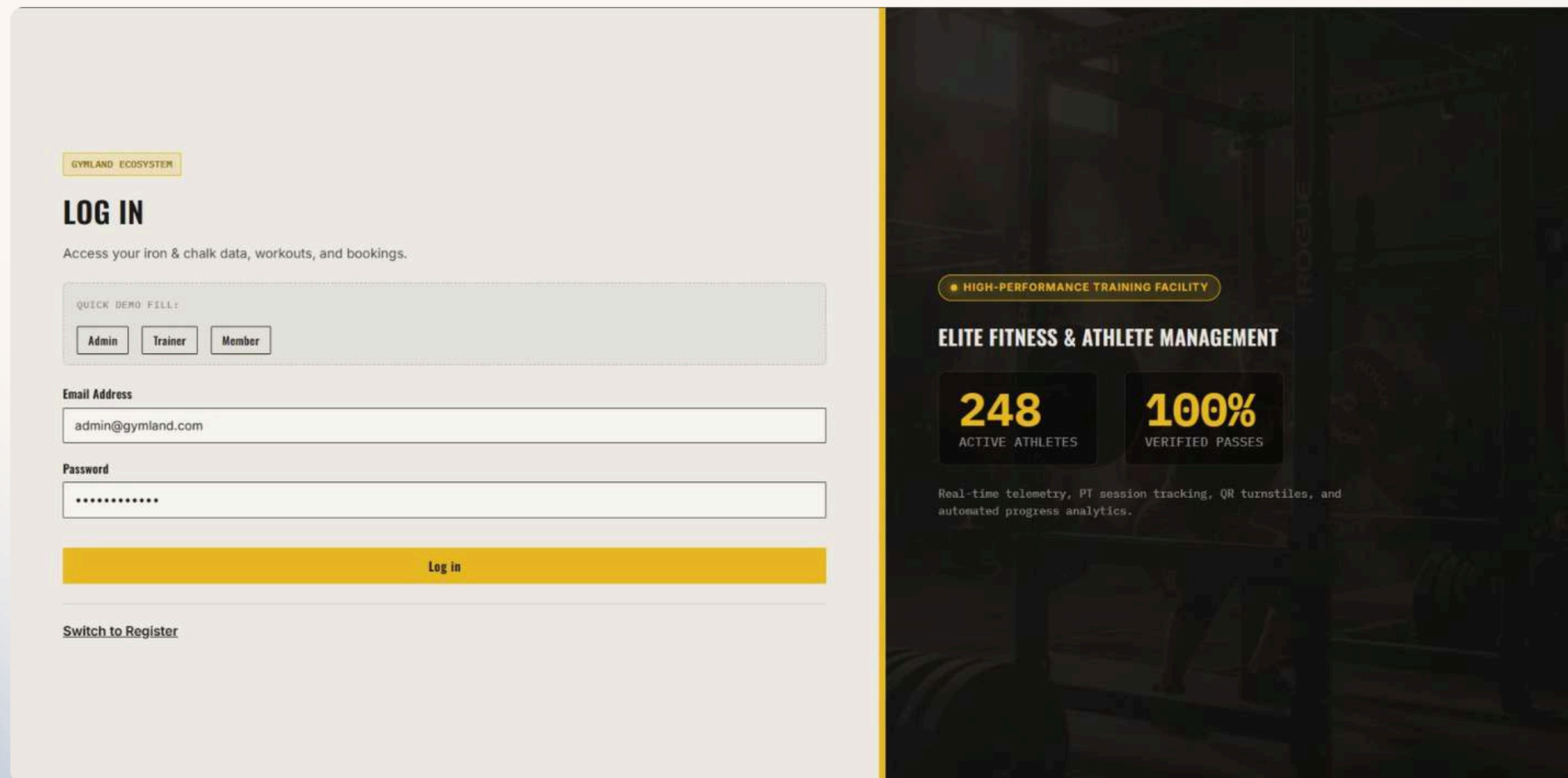
ASSIGNED MEMBER ROSTER (1)

MEMBER	CONTACT	PT CREDITS	MEDICAL & EMERGENCY	ACTION
Alex Johnson	member@gymland.com +1 555-4829	3 CREDITS	Med: Previous mild shoulder strain, cleared for lifting. Emerg: Sarah Johnson (+1 555-9988)	Log PT Session (-1)

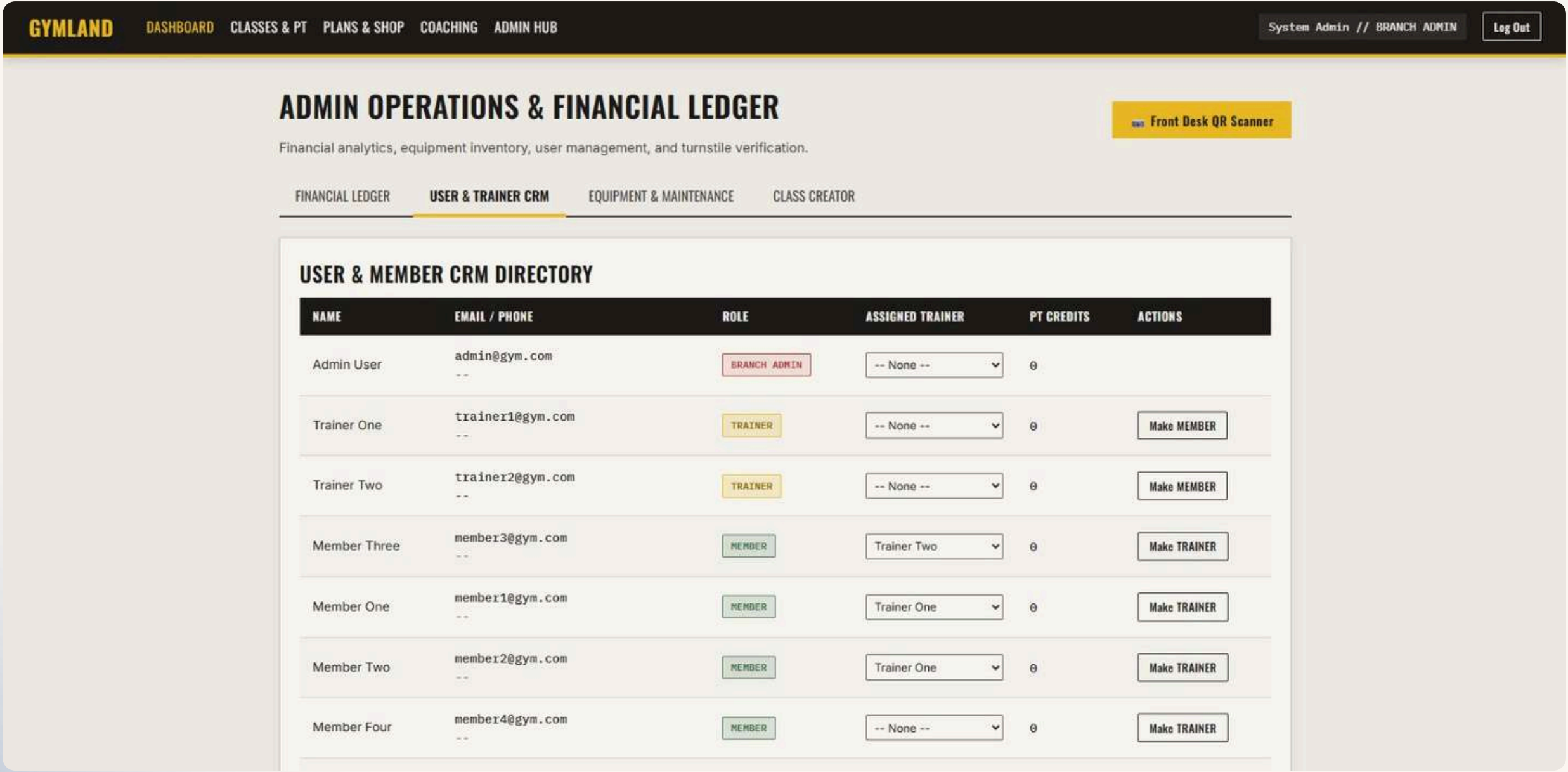
9. API TESTING & VALIDATION

- Modular REST API includes routes for authentication, users, classes, bookings, attendance, equipment, memberships, plans, payments, progress, reports, reviews, trainers, waitlists, and workout plans.
- Postman collection: `Gym-Management-API.postman_collection`
- Helper scripts: `generate_postman.js` and `test_auth.js`
- Reactive forms and mandatory-field validation are used for booking, checkout, equipment, and class creation.
- Protected routes use JWT verification and role checks before business operations execute.

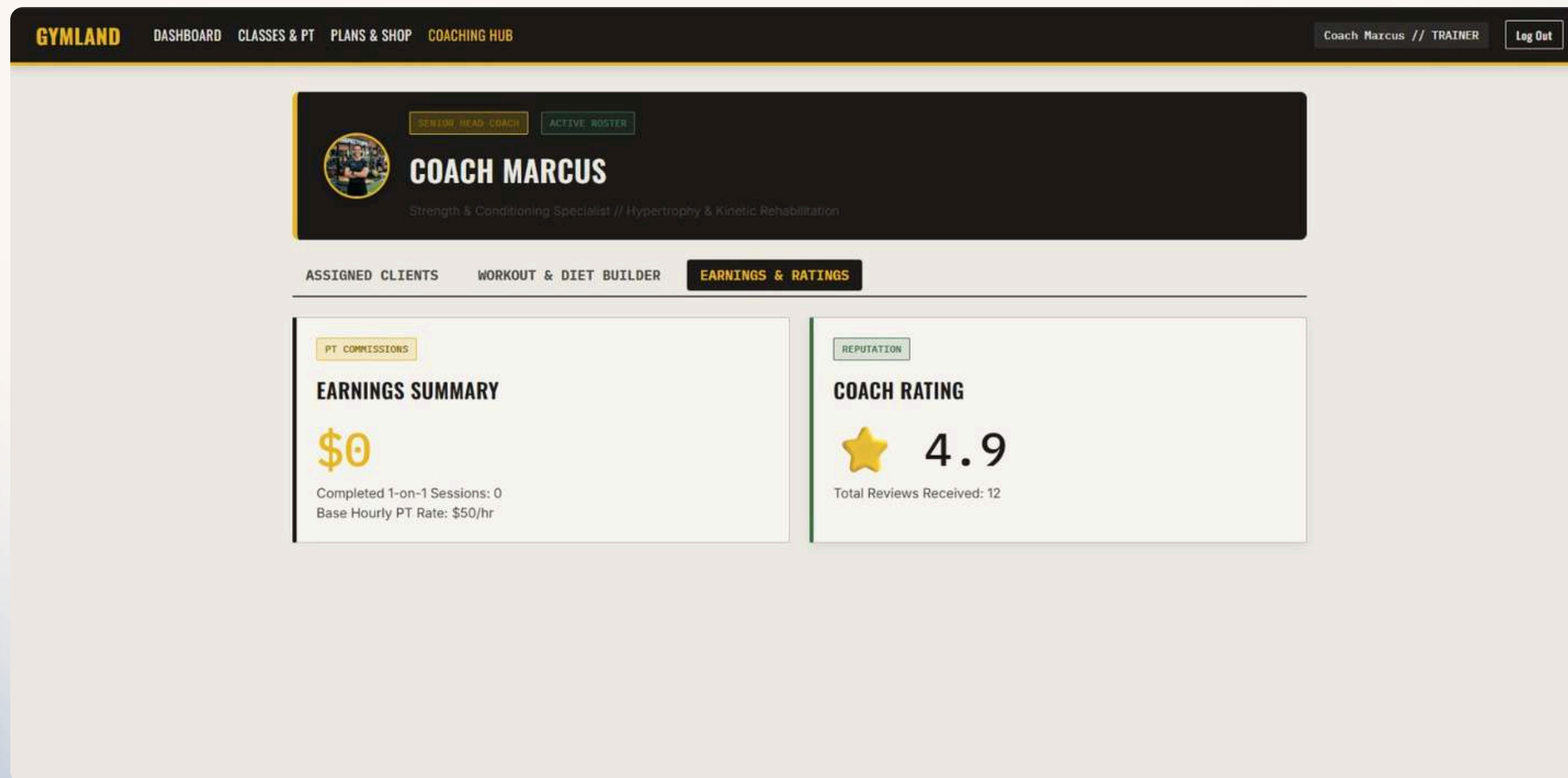
10. USER INTERFACE & FINAL OUTPUT



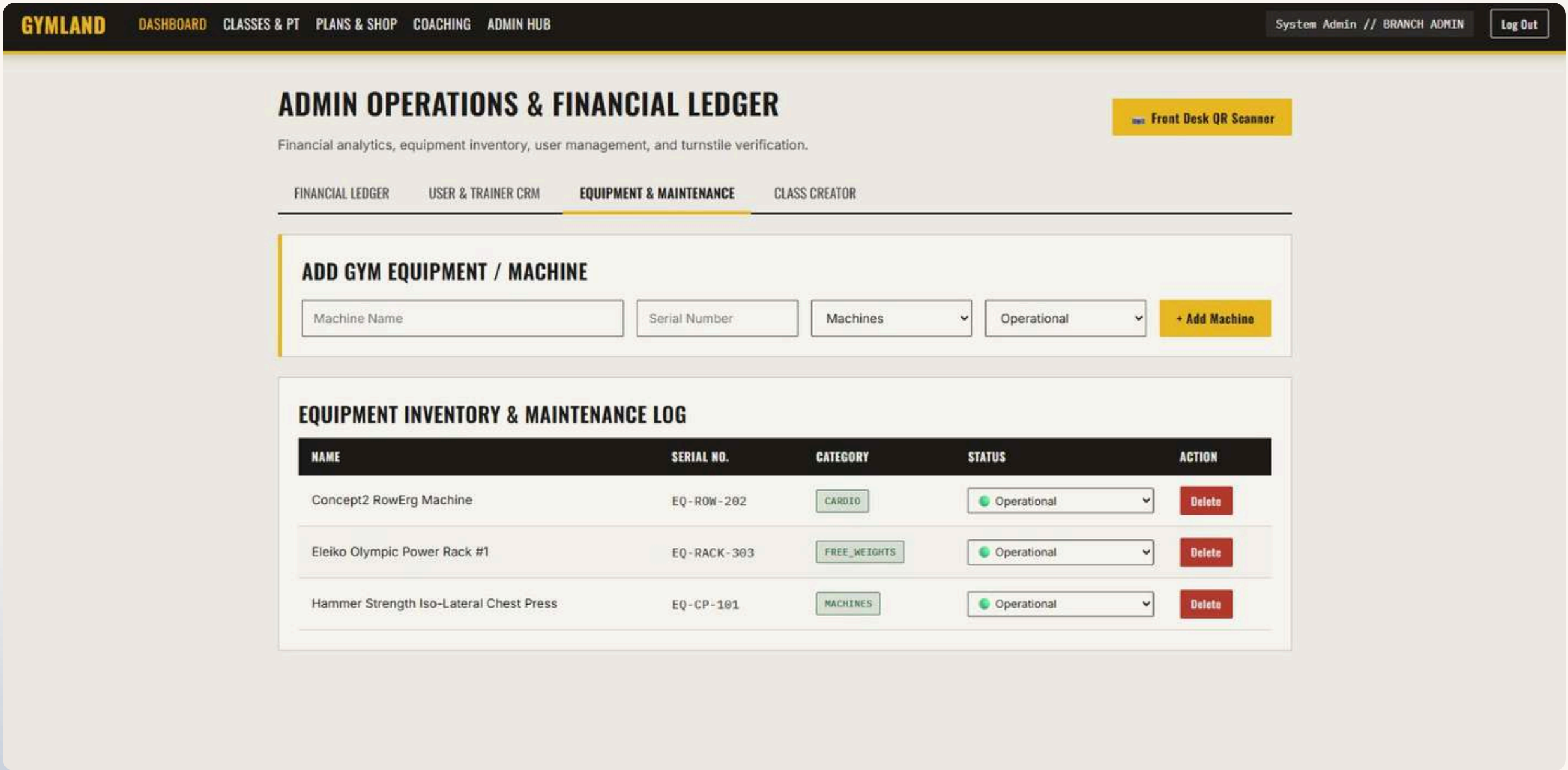
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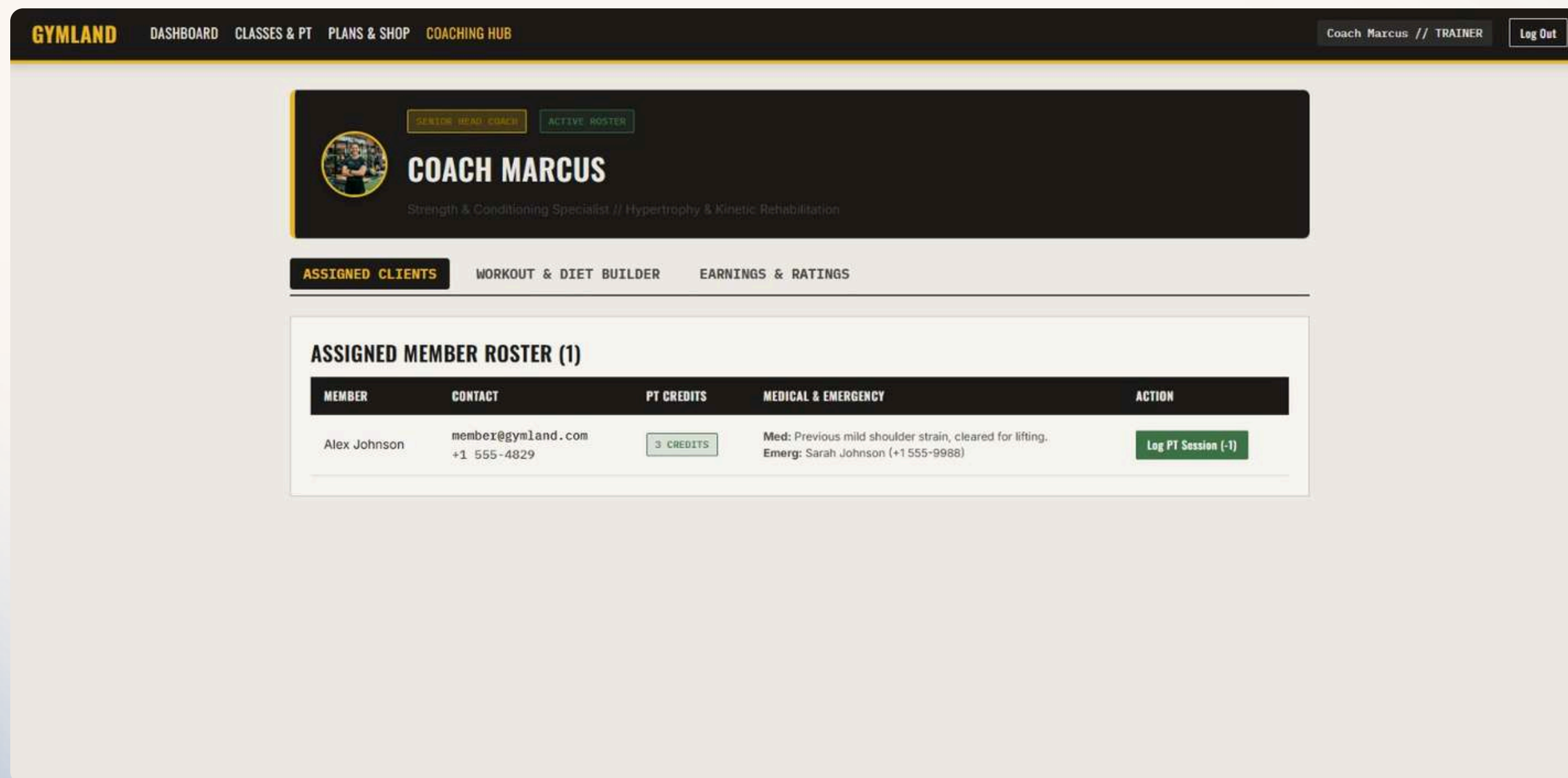
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11. SECURITY, VALIDATION & CODE QUALITY



Authentication

JWT-based authentication
Role-based route protection



Validation

Reactive forms
Mandatory fields
Dropdown/select checks



Architecture

Modular
routes/controllers/models
Centralized middleware



Reliability

Invalid IDs & bad input handled gracefully
Consistent API behavior

12. OUTCOME, LEARNINGS & CONCLUSION

- **Outcome:** a functional role-based gym management platform covering memberships, bookings, attendance, equipment maintenance, and coaching workflows.
- **Learning:** practical experience with Angular components, routing, services, reactive forms, Node.js/Express REST APIs, Mongoose modelling, and JWT authorization.
- **Conclusion:** GymLandT demonstrates how a modern MEAN-stack application can centralize gym operations into a scalable, secure, real-world workflow.

3

User roles

14+

Domain schemas

MEAN

Full-stack architecture

15+

API route groups

**THANK YOU
VERY MUCH**